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Biomonitoring of metals in fish and their parasites: an update

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ABSTRACT

Metal stress is one of the major concerns and persistent monitoring of the issue is the need of the hour. Urban domestic waste, agricultural runoff, industrial waste, and mining waste are anthropogenic sources of metals apart from natural sources. Specifically, the aquatic ecosystem including flora and fauna is under great threat due to metal discharges. Aquatic species, such as algae, fish, and associated parasites, are severely damaged due to metal pollution in turn affecting human beings through the food chain. To avoid this crisis, bioremediation is one of the best solutions. To perform bioremediation experiments, it is essential to monitor both flora and fauna near metal-rich areas. Hence, this review intends to summarize the developments in the biomonitoring of metals in fish and their parasites. The physiological, histopathological, biochemical, and molecular changes observed in both fish and their parasites grown in metal-rich water bodies were emphasized. Information on lethal doses and certain bioindicators were also highlighted in this review. This work will be useful to know the metal accumulation capacity of fish and their parasites, future bioremediation research using aquatic animals, and also helpful for fish industries.

ARTICLE HISTORY

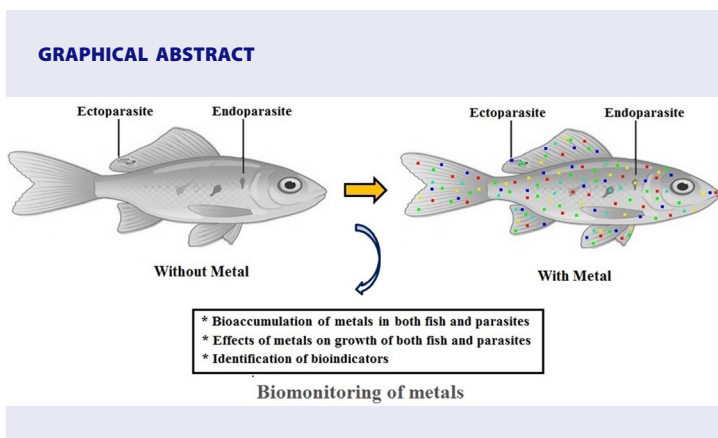
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Introduction

Undoubtedly our environment is under great stress due to natural and anthropogenic pollution and regular monitoring is therefore essential. Besides water and air pollution, metal pollution has become a major risk and the biota including human beings is being contaminated (Sures et al. 2017). Specifically, the physical condition and constancy of the aqueous biotope are at risk due to the alteration in the chemical composition of aquatic microhabitats (Hassan, Moharram, and El Helaly 2018). Metal pollution is one of the reasons for variations in the chemical composition of the aqueous biotope. Domestic waste, agricultural runoff, industrial waste, and mining waste are the major anthropogenic sources of metals apart from natural reasons such as weathering and volcanic eruption (Chandrasekhar, Keerthi Kumari, and Bramhachari 2022). The most dangerous impact of these metals is that they get incorporated into flora and fauna including fish and their parasites to make their way into the food chain (Kawade 2020).

Globally, various regions are accumulated with different metals which depend mostly on climatic conditions. Generally, metals are classified into two groups based on their usage (Kim et al. 2020). The first one includes biologically essential (Cu, Zn, Fe, Mn, etc.) and the other is nonessential or toxic metals (Cd, Pb, Hg, U, etc.). In addition, the responses of organisms to metals completely depend on the species (Mehana et al. 2020). The degree of metal pollution can be measured based on physical, chemical, and biological parameters. The absolute information about the ecosystem's stability can be provided by bioindicators such as aquatic plants including algae, microbial organisms, and aquatic animals including fish and their parasites (Vidal-Martínez 2007). Though some species exhibited tolerance to high metallic contents, most of them showed growth defects. Hence, initial biomonitoring is the best solution to know the metal toxicity

level which is a rapidly growing research area in recent times (Brazova et al. 2012). Further, bioremediation for metal cleanup is a popular mechanism since the olden days but the term molecular bioremediation is an advanced one where we can see and identify the responsible factors for remediation (uptake of metals and their function) in different organisms (Al-Hasawi 2019). Several molecular factors are involved in metal uptake and their function in plant and animal cells analyzed using advanced microarray or next-generation sequencing (NGS) or transcriptome sequencing techniques.

Accumulation of metals even below permissible levels creates health hazards in aquatic species including fish and their parasites. It is a well-known fact that fish is considered one of the important sources of food all over the world (Vankara and Chikkam 2017). Though global fish farming has reached a substantial level, regrettably there is a severe threat from pathogens such as viruses, bacteria, parasites, etc. (Modi and Vankara 2021). Several reports proved that fish and their parasites (both ecto and endo) act as main accumulators of both essential and non-essential metals which in turn depend on the species (Hassan, Al-Zanbagi, and Al-Nabati 2016). The accumulated metals affect the growth and development of both fish and their parasites.

Fish take up the metals and accumulate in the gills, intestine, liver, blood plasma, muscles including other organs of the body (Mziray and Kimirei 2016). Fish yield penalty is one of the major concerns and fishing industries need to take precautions to avoid the crisis. Moreover, fish and parasites are drawing the attention of scientists as potential indicators of environmental quality and remediators of pollution in a variety of ways (Najm and Fakhari 2015; Sures 2004). Particularly, the deteriorating state of the environment in aquatic resources can be reflected by an increased degree of parasites (Vankara et al. 2022). The present review mainly focused on the effects of various metals on fish growth patterns observed through morphological, histopathological, physiological, and biochemical methods. In addition, lethal doses and the accumulation capacity of metals in various fish and their parasites were emphasized. Further, fish and parasite interactions and figuring out the bio-indicators for certain metals were highlighted. Moreover, a number of toxicological studies were conducted by several researchers with different metals using fish and their parasites (Islam et al. 2020). Overall in this review, the biomonitoring of metals through fish and their parasites has been highlighted in detail. Figure 1 explains the biomonitoring and bioaccumulation studies of metals using fish and their parasites.

This updated knowledge will certainly be helpful to identify the bio-indicators, as well future research on metal's effects on fish and their parasites. Also, this information will be useful for fish industries to take precautionary measures.

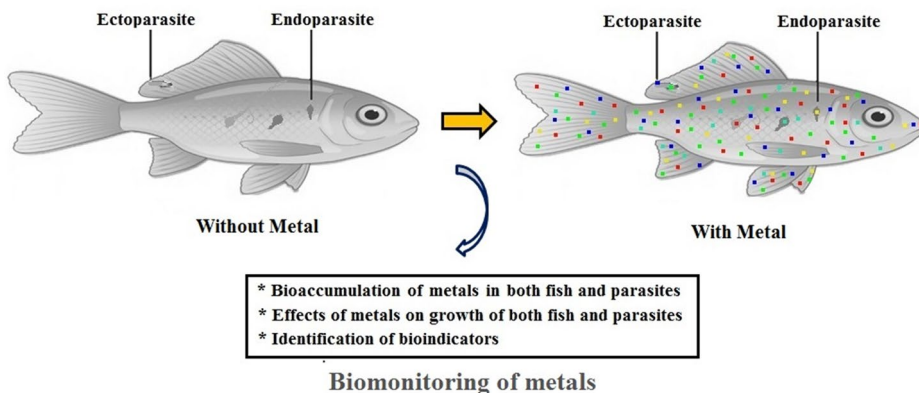


Figure 1. Schematic representation of the present study.

Sources and types of metals

The activities of the modern world polluted the environment and right from the beginning both flora and fauna are being continuously confronted with metal contamination (Nisha, Sekar, and Chandran 2016). Several species such as phytoplanktons including algae, macrophytes, microbial organisms, invertebrates, fish, and their parasites can be extensively used as bioindicators for metal pollution (Burger 2006). Metal pollution is increasing day by day due to both natural and anthropogenic activities. Apart from natural weathering and volcanic eruption, domestic wastes, sewage sludge, industrial effluents, mining waste, etc. are the main sources of metals (Brazova et al. 2012). Naturally occurring weathering process will take a long time and volcanic eruption is not very frequent. Specifically, industries such as metal processing, mining, chemical, etc. are the main sources of soil and water pollution. In addition, urban domestic waste is the major source of aquatic metal pollution (Reddy, Jagadeshwarlu, and Sunitha Devi 2016). Agricultural runoff is another source of metals which flow into the water bodies and contaminate the system. Figure 2 illustrates the anthropogenic reasons for metal contamination apart from natural reasons.

Scientifically, metals are classified based on relative densities, atomic weights, and atomic numbers. But based on their usage they are classified into essential and non-essential groups. Normally, nonessential group is considered toxic groups (Afshan et al. 2014). Among two kinds of metals, essential metals such as iron (Fe), copper (Cu), manganese (Mn), zinc (Zn), etc. are nutritionally essential for most living organisms. Moreover, certain essential metals when crossing their critical level may cause metabolic disorders and growth inhibition in most of the species (Delahaut et al. 2020). In addition, nonessential metals such as mercury (Hg), cadmium (Cd), chromium (Cr), lead (Pb), etc. are not used due to their

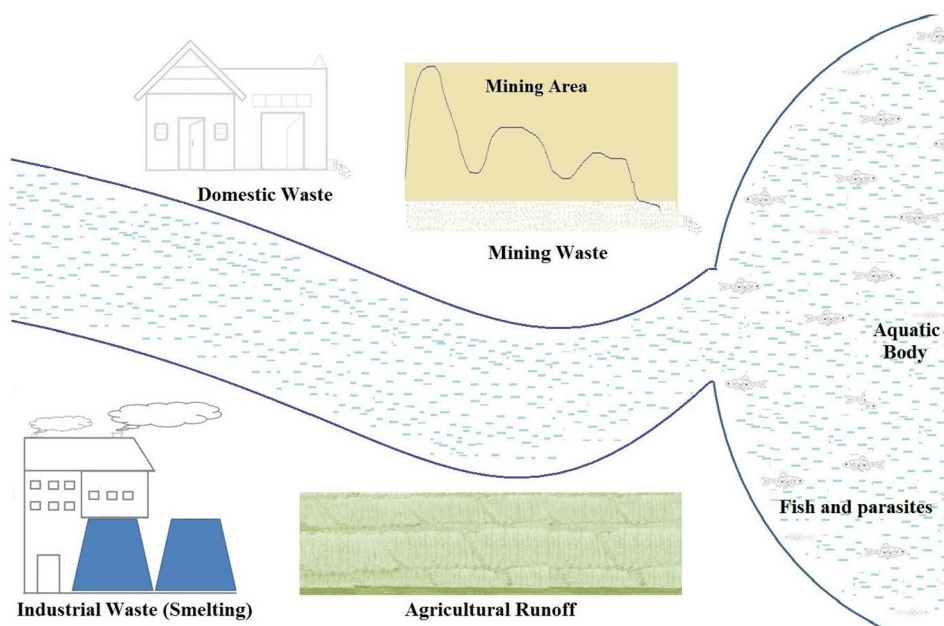


Figure 2. Anthropogenic reasons for metal discharges.

noxious nature and they may become a hazard to growth and development of plants, microbial organisms, and animals including human beings and fish.

Metal discharge in the aquatic bodies is considered to be the most important form of pollutants due to their toxicity and tendency to be accumulated by various aquatic organisms including fish which in turn affects human health (Brix et al. 2004). In certain fishes and their parasites, these metals are severely affected the growth, particularly by damaging the gills, liver, kidney, muscles, blood plasma, and intestine areas, and finally lead to death (Barson et al. 2014; Sures 2003). Several diseases were noticed in various fishes due to metal pollution. In certain cases, the appearance of parasites is an indication of metal pollution. Because of certain metals, parasite growth and development were affected in the host tissues (Al-Hasawi 2019; Jirsa et al. 2008; Shahat et al. 2011). Specifically, the deteriorating state of the environment in aquatic resources can be reflected by an increased degree of parasites. Knowledge of the toxicity of various metals on fish and their parasites is known for several years but the molecular research in this area focused recently.

Biomonitoring of metals using fish and their parasites

The aquatic environment receives various metal pollutants resulting from various anthropogenic activities especially in recent times (Erasmus et al. 2020). Specifically, non-essential metals such as Hg, Cd, Pb, U, etc. in

water bodies near metal contaminated areas have a tendency to be accumulated in fish and their parasites (Akinsanya and Kuton 2016). A number of biomonitoring and bioremediation research works were carried out using plants, algae, microbial organisms, invertebrates, fish, and parasites worldwide (Chandrasekhar, Keerthi Kumari, and Bramhachari 2022; Delahaut et al. 2020). But the biomonitoring and bioremediation studies of metal-rich areas using fish and their parasites were comparatively less. Some international laboratories emphasized the effects of metals on fish and parasite interactions and identified them as bioindicators by choosing various metal-rich water bodies.

The lethal effects of lead on European eel (*Anguilla anguilla*) and their parasites, that is, acanthocephalans and nematodes were emphasized by Sures and colleagues in 1994 (Sures, Taraschewski, and Jackwerth 1994) and Zimmerman and colleagues in 1997 (Zimmerman, Sures, and Taraschewski 1997). Similarly, Sures, Siddall, and Taraschewski (1999) and Sures and Siddall (1999) highlighted the role of lead and cadmium on fish (*Leuciscus cephalus*) and their parasites such as *Ascaris suum* (nematoda), *Fasciola hepatica* (digenea), *Pomphorhynchus laevis* (acanthocephalan) including certain cestodes and identified these organisms as bioindicators for metal pollution (Table 1). These results also proved that biomonitoring is necessary to check the metal contamination level using fish and their parasites. Sures (2001) examined various fish parasites and elucidate the importance of parasites as bioindicators of metals in the aquatic ecosystem. Severe toxicity of copper was noticed in bull trout (*Salvelinus confluentus*) and rainbow trout (*Oncorhynchus mykiss*) by Hansen, Lipton, and Welsh (2002). At 220mg/l water hardness, bull trout showed less sensitivity than rainbow trout. Turcekov, Hanzelova, and Spakulova (2002) analyzed the lethal doses of As, Cd, Cu, Pb, and Zn using organs of *Perca fluviatilis* (European perch) and their anthocephlan and tapeworm parasite tissues.

Thophon et al. (2003) observed the histopathological changes in white seabass, *Lates calcarifer* when treated with cadmium. Sures and Reimann (2003) observed that metals are accumulated several times more in the parasites than in the muscles and liver of their fish host. Similarly, Schludermann et al. (2003) found fish macroparasites as metal indicators by choosing a metal-polluted aquatic system. The ion regulatory system was disrupted in lead-treated rainbow trout compared to normal fish (Rogers, Richards, and Wood 2003). Indian major carp, *Labeo rohita* treated with chromium altered the biochemical composition (Vutukuru 2003). Tilapia (*Oreochromis mossambicus*) exposed to arsenic waste in the blackfoot disease (BFD) area in Taiwan found variations in biochemical pathways (Liao et al. 2003). The highest BCF (bioconcentration factor) was found in the intestine and high

Table 1. Biomonitoring of metals in fish and their parasites.

S. No.	Biomonitoring of metals in fish and their parasites	Metals	Authors	Year
1.	<i>Anguilla anguilla</i> with <i>Paratenuisentis ambiguous</i> (Acanthocephala), <i>Anguillicola crassus</i> (Nematodes)	Pb	Sures, Taraschewski, and Jackwerth	1994
2.	Eel with specific endoparasites <i>Anguillicola crassus</i> (Nematoda) and <i>paratenuisentis ambiguous</i> (Acanthocephala)	Pb	Zimmerman, Sures, and Taraschewski	1997
3.	<i>Leuciscus cephalus</i> and <i>Pomphorhynchus laevis</i>	Pb	Sures and Siddal	1999
4.	Perch (<i>Perca</i>) and its endoparasites	As, Cd, Cu, Pb, and Zn	Turcekov, Hanzelova, and Spakulova	2002
5.	<i>Salvelinus confluentus</i>	Cu	Hansen,	2002
6.	<i>Oncorhynchus mykiss</i>		Lipton, and Welsh	
7.	<i>Lates calcarifer</i>	Cd	Thophon et al.	2003
8.	<i>Cirrhinus mrigala</i>	Ni and Cr	Virk and Sharma	2003
8.	<i>Notothenia coriiceps</i> and <i>Aspersentis megarhynchus</i> (Acanthocephala)	Al, Ag, As, Ba, Cd, Co, Cr, Cu, Fe, Mg, Mn, Ni, Pb and Sr	Sures and Reimann	2003
9.	<i>Labeo rohita</i>	Cr	Vutukuru	2003
10.	<i>Oncorhynchus mykiss</i>	Ni	Pane, Richards, and Wood	2003
11.	<i>Oncorhynchus mykiss</i>	Pb	Rogers, Richards, and Wood	2003
12.	<i>Oreochromis mossambicus</i>	As	Liao et al.	2003
13.	<i>Poecilia reticulata</i>	Cd	Yilmaz, Gül, and Karakose	2004
14.	<i>Oncorhynchus mykiss</i>	Ni	Brix et al.	2004
15.	<i>Channa punctatus</i>	Hg	Pandey et al.	2005
16.	<i>Gasterosteus aculeatus</i> and <i>Rutilus rutilus</i> and <i>Oncorhynchus mykiss</i> and <i>Perca fluviatilis</i> and <i>Leuciscus leuciscus</i>	Cr	Svecevičius	2006
17.	<i>Catla catla</i> <i>Cirrhina mrigala</i> <i>Labeo rohita</i>	Fe and Ni	Javed and Abdullah	2006
18.	<i>Trachurus trachurus</i> Other fishes	Fe, Zn, Ni,Pb, Cr,Cu,Cd,Mn and Se	Demirezen and Uruc	2006
19.	<i>Carcharhinus dussumieri</i> and parasites	Pb and Cd	Malek et al.	2007
20.	<i>Labeo rohita</i>	As	Vutukuru et al.	2007
21.	<i>Clarias batrachus</i>	As	Bhattacharya and Bhattacharya	2007
22.	<i>Catla catla</i>	Cd	Sobha et al.	2007
23.	<i>Chondrostoma nasus</i> its intestinal parasite <i>Caryophyllaeus laticeps</i>	Cu, Zn, Pb and Cd	Jirsa et al.	2008
24.	<i>Anabas testudineus</i>	As and Hg	Akter et al.	2008
25.	<i>Danio rerio</i>	As	Liu, Gentles, and Theodorakis	2008
26.	<i>Channa punctatus</i>		Murugan et al.	2008
27.	<i>Carassius auratus</i> <i>Xiphophorus helleri</i>	Cu	James et al.	2008
28.	<i>Channa punctatus</i>	Cr	Mishra and Mohanty	2008
29.	<i>Liza aurata</i>	Cd and Cr	Pazouki, Abtahi, and Rezaei	2009
30.	<i>Clarias gariepinus</i>	As	Hamdi et al.	2009
31.	<i>Leporinus macrocephalu</i> <i>Prochilodus vimbooides</i>	Cu and Cd	Gomes et al.	2009
32.	<i>Pomatoschistus microps</i>	Hg and Cu	Vieira et al.	2009
33.	<i>Pagrus major</i>	Cd	Cao et al.	2009
34.	<i>Cirrhinus mrigala</i>	Cr and Ni	Palaniappan and Karthikeyan	2009

(Continued)

Table 1. Continued.

S. No.	Biomonitoring of metals in fish and their parasites	Metals	Authors	Year
35.	<i>Abramis brama</i> <i>Rutilus rutilus</i> <i>Cyprinus carpio</i>	Ni	Skibniewska et al.	2009
36.	<i>Catla catla</i>	As	Kavitha et al.	2010
37.	<i>Lethrinus nebulosus</i> , <i>Caranx sexfaciatus</i> , <i>Carans melampygus</i> and <i>Lethrinus mahsena</i>	Zn, Cu, As, Cd, Hg and Pb	Ali, Elazein, and Alian	2011
38.	<i>Oreochromis niloticus</i> niloticus and <i>Clarias gariepinus</i> Acanthocephalans, nematodes and trematodes	Cu, Cd and Pb	Shahat et al.	2011
39.	<i>Catla catla</i>	As	Lavanya et al.	2011
40.	<i>Esox Lucius</i>	Cd, Cu, Pb and Zn	Ebrahimi Sirizi, Sakizadeh, and Esmaili Sari	2012
41.	<i>Perca fluviatilis</i> and its intestinal parasite <i>Acanthocephalus lucii</i>	Zn, Mn, Cu and Cd	Jankovská et al.	2012
42.	Perch and its parasites	As, Cd, Cr, Cu, Hg, Mn, Ni, Pb and Zn	Brazova et al.	2012
43.	<i>Channa punctatus</i>	As	Das et al.	2012
44.	<i>Channa gachua</i>	Hg and Cu	Deore and Wagh	2012
45.	<i>Oreochromis mossambica</i> , <i>Cyprinus carpio</i> , <i>Salmophasia horai</i>	U	Amarnath Reddy et al.	2013
46.	<i>Barbus barbus</i> and parasites <i>Pomphorhynchus laevis</i> and <i>Eustrongylides</i>	As, Cd, Co, Cu, Fe, Mn, Pb, Se, Sn, V and Zn	Nachev, Schertzinger, and Sures	2013
47.	<i>Cyprinus carpio</i>	As	Kovendan and Vincent	2013
48.	<i>Oreochromis mossambicus</i>	As	Ahmed et al.	2013
49.	<i>Sparus aurata</i>	Cd	Guardiola et al.	2013
50.	<i>Danio rario</i> and <i>Ctenopharyngodon idella</i> and <i>Gobiocypris rarus</i>	Cu, Hg, Zn and Cd	Wang et al.	2013
51.	<i>Rutilus kutum</i>	Pb and Cu	Gharedaashi, Imanpoor, and Taghizadeh	2013
52.	<i>Cyprinus carpio</i> (Common carp)	Pb, Cd, Ni and Cr	Afshan et al.	2014
53.	<i>Clarias gariepinus</i> and helminths	Cd, Cr, Cu, Fe, Ni, Pb and Zn	Barson et al.	2014
54.	<i>Labeo rohita</i> <i>Cirrhinus mrigala</i> <i>Catla catla</i> <i>Ctenopharyngodon idella</i>	Ni, Zn and As	Kousar and Javed	2014
55.	Monogeneans, Digeneans and Crustaceans of 360 specimens of three fish species	Zn, Se, Fe, Cr and Ni	Bayoumy, Abou-El-dobal, and Hassanain	2015
56.	<i>Rasbora sumatrana</i>	Cu, Pb, Fe, Al, Mn, Ni, Zn and Cd	Shuhaimi-Othman et al.	2015
57.	<i>Poecilia reticulata</i> <i>Clarius batrachus</i>	Cr	Johnson and Radhakrishnan	2015
58.	<i>Oryzias latipes</i>	Cr	Li et al.	2015
59.	<i>Synodontis clarias</i> and <i>Chrysichthys nigrodigitatus</i> and parasites	Fe, Mn, Zn, Pb and Cd	Akinsanya and Kuton	2016
60.	<i>Epinephelus summana</i> Nematode helminthes	Ca, K, Na, Fe, Pb, Zn, Mg and Cu	Hassan, Al-Zanbagi, and Al-Nabati	2016
61.	<i>Siganus sutor</i> , <i>Lethrinus harak</i> , and <i>Rastrelliger kanagurta</i>	Ag, As, Co, Cr, Mn, Ni, Se, Al, Cd, Cu, Fe, Pb and Zn	Mziray and Kimirei	2016
62.	<i>Sarotherodon mossambica</i>	Cu	Reddy, Jagadeshwarlu, and Sunitha Devi	2016

(Continued)

Table 1. Continued.

S. No.	Biomonitoring of metals in fish and their parasites	Metals	Authors	Year
63.	<i>Oreochromis sp.</i>	Cu, Zn and Cd	Aldoghachi et al.	2016
64.	<i>Danio rario</i>	Cr	Nisha, Sekar, and Chandran	2016
65.	<i>Labeo rohita</i>	Cr	Bakshi	2016
66.	<i>Oncorhynchus mykiss</i>	Pb	Alsop et al.	2016
67.	<i>Lethrinus mahsena</i> Helminths	As, Fe, Zn, Pb, Cu and Cd	Hassan, Moharram, and El Helaly	2018
68.	<i>Sclerocollum saudii</i>	Cd and Pb	Hassanine et al.	2018
69.	<i>Clarias gariepinus</i>	Pb	Samuel et al.	2018
70.	<i>Clarias batrachus</i>	Pb	Pandit, Kumari, and Garhwal	2018
71.	Siganid fish <i>Siganus rivulatus</i> and intestinal helminth parasites	Cd and Pb	Al-Hasawi	2019
72.	<i>Channa striata</i> , <i>Labeo rohita</i> and <i>Catla catla</i>	Ni, Pb, Fe, Cu, Zn, Cd, Cr and Co	Kumar et al.	2020
73.	Fish helminth parasites	Cd, Cr, Cu, Ni, Pb, Pt and Zn	Erasmus et al.	2020
74.	<i>Cyprinus carpio</i>	Cd	Delahaut et al.	2020
75.	<i>Pangasianodon hypophthalmus</i>	Cr	Islam et al.	2020
76.	<i>Channa gachua</i>	Ni	Kawade	2020
77.	<i>Danio rerio</i>	Pb	Kim et al.	2020

accumulation was noticed in muscles and viscera. Hence, consuming fish without viscera may reduce the health risk. Waterborne nickel worked as a respiratory toxicant in rainbow trout and exhibited less impact on the ion regulatory system (Pane, Richards, and Wood 2003). Virk and Sharma (2003) assessed the biochemical composition using muscle samples of nickel and chromium-treated *Cirrhinus mrigala* after exposure and the subsequent withdrawal of the metals. They analyzed the toxicity levels and estimated the LC_{50} values for both metals. Similar rainbow trout exhibited high sensitivity to nickel when treated with both acute and chronic doses (Brix et al. 2004). 20.8 mg/l was the acute median lethal concentration for nickel and the 24-h gill median lethal accumulation was 666 nmol/g wet weight. They also concluded that nickel is not an ion regulatory toxicant in long-term exposures. Yilmaz, Gül, and Karakose (2004) noticed the behavioral changes in guppy fish (*Poecilia reticulata*) when treated with cadmium chloride. They found imbalanced swimming, difficulty in breathing, capsizing, etc. in cadmium chloride-treated fish. Air-breathing *Channa punctatus* was treated with mercuric chloride and malathion individually and found 1.15 ppm and 6.61 ppm as the 96 h LC_{50} values in SPSS method (Pandey et al. 2005). The mortality rate was enhanced with increased dose and exposure timings.

Three fishes namely *Catla catla*, *Labeo rohita* and *Cirrhina mrigala* were treated with iron and nickel independently to know the 96 h LC_{50} values by Javed and Abdullah (2006). The lethal concentration of iron ranged from 72.20 mg/l to 164.65 mg/l and *C. mrigala* showed high

sensitivity. The lethal concentration of nickel fluctuated between 20.20 mg/l to 50.11 mg/l and *C. catla* exhibited more sensitivity to this metal. Svecevicus (2006) noticed the toxic nature of hexavalent chromium when treated with European freshwater fish. The authors mentioned that the metal sensitivity levels will also depend on the region. Demirezen and Uruc (2006) monitored metals using fish and other species. Malek et al. (2007) identified certain parasites of sharks as metal bioindicators. Oxidative stress i.e. lipid peroxidation and induction of hydrogen peroxide was noticed by arsenic treatment in Indian catfish, *Clarias batrachus* (Bhattacharya and Bhattacharya 2007). Also, enhancement of activities of antioxidant enzymes such as superoxide dismutase (SOD), glutathione peroxidase (GPX), and glutathione reductase (GR) was noticed in arsenic-treated fish. Biochemical changes were noticed in cadmium-treated freshwater fish *Catla catla* by Sobha et al. (2007). Altered serum amino-transferases activity was noticed in *Labeo rohita* when treated with arsenic and chromium (Vutukuru et al. 2007). The intestine of common nase fish and their parasites found with metals by Jirsa et al. (2008). Arsenic and mercury-treated freshwater climbing perch (*Anabas testudineus*) exhibited erratic movement, loss of equilibrium, lateral swimming, secretion of excessive mucus, rapid movement of the operculum, jumping out of the test media, etc. (Akter et al. 2008). This abnormal behavior is due to the direct contact of metals, metabolic alterations, and neurotoxic effects. Also, they found 18.211 ppm and 0.606 ppm as LC_{50} values of arsenic and mercury respectively. James et al. (2008) determined the 96 h LC_{50} values of copper for ornamental fish, that is, *Carassius auratus* (0.30 ppm) and *Xiphophorus helleri* (0.36 ppm). They also found that copper accumulated more in muscle and gonad tissue at high doses. Mishra and Mohanty (2008) found the behavioral and histopathological changes in hexavalent chromium-treated *Channa punctatus*. Specifically, erratic swimming along with epithelial hyperplasia, epithelial necrosis, epithelial lifting, lamellar fusion, oedema, aneurism, etc. were noticed in metal-treated fish (Table 1). Liu, Gentles, and Theodorakis (2008) studied arsenate and perchlorate toxicity using zebrafish.

Cao et al. (2009) found that red sea bream (*Pagrus major*) embryos were more sensitive to cadmium than larvae. Also, they found that 24 and 48 h LC_{50} values of cadmium for embryos were 9.8 and 6.6 mg/l. In addition, 24, 48, 72, and 96 h LC_{50} values for larvae were 18.9, 16.2, 8.0 and 5.6 mg/l, respectively. Freshwater fingerlings (*Cirrhinus mrigala*) were monitored for chromium and nickel accumulation in contaminated water bodies (Palaniappan and Karthikeyan 2009). Fish kidneys were severely damaged by chromium contamination. Also, they noticed that chromium accumulated more along with nickel binary mixtures than those of individual metals. Pazouki, Abtahi, and Rezaei (2009) estimated the Cd and

Cr accumulation levels in the muscle and skin of golden grey mullet fish. Gomes et al. (2009) identified the LC_{50} values of copper (0.047 mg/l and 0.090 mg/l) and cadmium (3.16 mg/l and 7.42 mg/l) for curimata (*Prochilodus vimbooides*) and Piauçu (*Leporinus macrocephalus*) respectively. Curimata was more sensitive to these metals than the other fish. Three freshwater fishes i.e. *Cyprinus carpio*, *Rutilus rutilus*, and *Abramis brama* were treated with nickel and found variations in accumulation capacity by Skibniewska et al. (2009). Vieira et al. (2009) treated estuarine fish *Pomatoschistus microps* with copper and mercury and found 568 µg/l and 62 µg/l as LC_{50} values. Swimming performance was assessed using a device specially developed for epibenthic fish (SPEDE) for treated and control fish. Lipid peroxidation (LPO) and enzymes (SOD, CAT, GR, GST, AChE, and LDH) were measured using both treated and untreated fish samples. Arsenate-exposed *Catla catla* resulted in alterations in biochemical, hematological, and liver transaminases activities (Kavitha et al. 2010). They found 43.78 mg/l as a lethal dose (96 h LC_{50}) for this fish and found decreased levels of white blood cells, red blood cells, hemoglobin, hematocrit, plasma protein and glucose, and liver aspartate and alanine aminotransferases (AST and ALT). In contrast, mean cell hemoglobin (MCH), mean cell hemoglobin concentration (MCHC), and corpuscular indices like mean cell volume (MCV) increased. Ali, Elazein, and Alian (2011) found metals such as Zn, Cu, As, Cd, Hg, and Pb in four fish species, that is, *Lethrinus nebulous*, *Caranx sexfaciatus*, *Carans melampygus*, and *Lethrinus mahsena* in Red Sea-Jeddah Coast (Table 1). Lavanya et al. (2011) and Shahat et al. (2011) monitored the metals in both fish and their parasites.

Ebrahimi Sirizi and colleagues (Ebrahimi Sirizi, Sakizadeh, and Esmaili Sari 2012) surveyed the Cd, Pb, Cu, and Zn contamination levels in muscle tissue of northern pike fish. Das et al. (2012) found that 100 ppm arsenic kills the teleost *Channa punctatus* within 18 to 20 h. This work is solid proof for biological indication and these teleosts were proved as biological indicators for arsenic pollution. Effect of mercuric chloride and copper chloride in the liver of freshwater teleost, *Channa gachua* was evaluated by Deore and Wagh (2012). With this fish, 1.0625 ppm of mercuric chloride and 1.4202 ppm of copper chloride were found as lethal doses. Histopathological evidences such as degeneration of nuclei, swellings, vacuolation in cytoplasm and stroma, necrosis, rupture of blood sinusoids, alteration in the shape of hepatocytes, etc. proved the fish damage. Median lethal concentrations of copper sulphate and lead nitrate were 2.944 ppm and 315.841 ppm for Caspian sea kutum (*Rutilus frisii kutum*) noticed by Gharedaashi and colleagues (Gharedaashi, Imanpoor, and Taghizadeh 2013). Rapid opercular movement and frequent gulping of air were noticed when treated with these compounds. Copper proved

more lethal than lead in this fish which is proved by few researchers. Ahmed et al. (2013) found that 28.22 ppm of Arsenic is the median lethal concentration for freshwater fish, tilapia (*Oreochromis mossambicus*). Arsenic toxicity damages the gills and liver severely and alters fish behavior. Gills were characterized by epithelial hyperplasia, lamellar fusion, aneurism, desquamation and necrosis. Liver tissue exhibited focal lymphocytic and macrophage infiltration, shrinkage of hepatocytes, focal necrosis, nuclear hypertrophy, etc. Amarnath Reddy et al. (2013) found 30 species of freshwater fishes in Uranium mining areas which including *Oreochromis mossambica*, *Cyprinus carpio*, *Salmophasia horai*, etc. and survived with this toxic metal. Nachev and colleagues (Nachev, Schertzinger, and Sures 2013) estimated the accumulation of As, Cd, Co, Cu, Fe, Mn, Pb, Se, Sn, V, and Zn in the fish *Barbus barbus* and its parasites *Pomphorhynchus laevis* and *Eustrongylides*. Vaseem and Banerjee (2013) identified *Labeo rohita* fish as a bioindicator for some metals in Ganga River through biomonitoring work. Kovendan and Vincent (2013), Wang et al. (2013) and Guardiola et al. (2013) monitored various metals in different fishes.

Barson et al. (2014) highlighted the influence of metals on fish-parasite interaction by choosing helminth and African catfish. Afshan et al. (2014) and Kousar and Javed (2014) studied the effect of metal pollution on fish and found the toxicological effects. Bayoumy, Abou-El-dobal, and Hassanain (2015) proved that fish parasites as bioindicators for certain metals in a detailed study. Freshwater catfish (*Clarias batrachus*) were treated with chromium and found 35.50 ppm is LC_{50} for 96h (Johnson and Radhakrishnan 2015). Treated fish exhibited jerky movements, erratic swimming, mucus secretion, significant changes in opercular movement, etc. Japanese medaka (*Oryzias latipes*) treated with chromium exhibited deleterious effects (Li et al. 2015). Chromium accumulated in various tissues including the liver and induced lipid peroxidation. Also, chromium alters the functions of antioxidant enzymes such as catalase, superoxide dismutase, and acetylcholinesterase (Table-1). Shuhaimi-Othman et al. (2015) studied eight metal toxicity levels using freshwater fish. Nisha and colleagues (Nisha, Sekar, and Chandran 2016) noticed that 105.279 mg/l and 26.03 mg/l of trivalent and hexavalent chromium concentrations as lethal (96h LC_{50}) to zebra fish (*Danio rerio*). Erratic swimming, secretion of mucus, jerky movements, normal color change, etc. were found in chromium treatment. Accumulation of Pb, Mn, Fe, Zn, and Cd was noticed in *Synodontis clarias* and *Chrysichthys nigrodigitatus* and their parasites by Akinsanya and Kuton (2016). Juvenile hybrid red tilapia (*Oreochromis* sp.) was treated with metals (Cu, Cd, and Zn) exhibited toxic effects (Aldoghachi et al. 2016). The 96h LC_{50} values of Cu, Cd, and Zn were 0.45, 0.7, and 2.1 mg/l, respectively. All the three metals

differently accumulated in various parts of the fish. A higher concentration of all the metals leads to mortality. Alsop et al. (2016) used both waterborne and dietborne lead to treat rainbow trout, *Oncorhynchus mykiss*, and noticed the deleterious effects. Marginal effects were found in this fish with Pb treatment and also reduction of carbohydrates and lipids of oligochaete worms (*Lumbriculus variegatus*) located in this fish was observed. The results confirmed that dietborne lead is safer than waterborne lead for trout.

Bakshi (2016) found that 30.36 and 33.39 mg/l of chromium are the median lethal concentrations for *Labeo rohita* and *Heteroneustes fossilis*. Chromium altered the growth and development of the fish through deleterious effects. Reddy and colleagues (Reddy, Jagadeshwarlu, and Sunitha Devi 2016) identified the lethal dose (LC₅₀) of copper for *Sarotherodon mossambica*. Hematological and biochemical changes were noticed in copper-treated fish. Lipids, fats, and protein contents were reduced in metal polluted fishes. Activities of aspartate aminotransferase, alanine aminotransferase, and alkaline phosphatase were reduced. Similar metal monitoring works will be carried out by Mziray and Kimirei (2016) and Hassan, Al-Zanbagi, and Al-Nabati (2016) using fish and their parasites. Sures et al. (2017) mentioned that the parasites can be used either as accumulation indicators or effect indicators due to the diversity of ways in which they respond to anthropogenic pollution. Hassan, Moharram, and El Helaly (2018) identified the role of helminth parasites in the bioremediation process for certain metals in the tissues of emperor fish. Here fish accumulates arsenic, cadmium, copper, iron, lead, and zinc in various parts of the body including the liver, kidney, intestine, and muscles. Air-breathing *Clarias batrachus* exposed to a lethal dose of lead nitrate leads to a reduction in serum glucose, serum cholesterol, and serum urea levels (Pandit, Kumari, and Garhwal 2018). In contrast, serum protein levels were decreased with the same lethal dose. Samuel et al. (2018) noticed the poor feeding habit, erratic swimming, secretion of mucus, and eventually death in lead nitrate-treated *Clarias gariepinus*. Also, they found 284.189 mg/l is 96 h LC₅₀ with this fish. Hassanine et al. (2018) and Al-Hasawi (2019) doing similar works and reported that both fish and their parasites act as bioindicators for some metals in different parts of the world.

Recently, Kumar et al. (2020) performed the biomonitoring work using *Channa striata*, *Labeo rohita* and *Catla catla* for Ni, Pb, Fe, Cu, Zn, Cd, Cr, and Co metals and identified the accumulation levels. Mehana et al. (2020) highlighted the metal accumulation in Acanthocephalans parasite in the aquatic system. Keke et al. (2020) conducted the biomonitoring work in an afrotropical stream using helminth parasites of fish and identified them as bioindicators. A number of helminth parasites were found in various vertebrates including fish in platinum-rich regions and

identified as bioindicators (Erasmus et al. 2020). Delahaut et al. (2020) showed the accumulation and toxicity levels of Cd, Cu, and Zn in common carp (*Cyprinus carpio*) juveniles. Median lethal values (96h) of Cd, Cu, and Zn were $0.20 \pm 0.16 \mu\text{M}$, $0.77 \pm 0.03 \mu\text{M}$, and $29.89 \pm 9.03 \mu\text{M}$, respectively. Toxicity levels were high with cadmium and low with zinc in this fish. Islam et al. (2020) proved that 32.47 mg/l of chromium is the LC_{50} value (96h) for striped catfish *Pangasianodon hypophthalmus*. Hemoglobin and red blood cells were decreased with increased Cr concentrations. In contrast, white blood cells and glucose levels were increased with increased Cr doses. Significant increase in erythrocytic cellular abnormalities (ECA) and erythrocytic nuclear abnormalities (ENA) were found with increased chromium doses in this fish. Kawade (2020) noticed the alteration in liver activities in nickel-treated *Channa gachua* fish. Accumulation of blood cells in the central vein, hemorrhage in hepatocytes and connective tissue, shrinkage of central vein, hepatic tissue damage, etc. was noticed in nickel-treated fish liver. Determination of lead toxicity was analyzed using zebra fish in soft and hard water (Kim et al. 2020). Results revealed that lead is less toxic in hard water than in soft water. 96h LC_{50} values are around 28.62 mg/l and 18.3 mg/l in hard and soft waters. Recently, our laboratory emphasized the role of parasites of fishes in assessing the water quality in River Penna (Modi, Thummala, and Vankara 2021). But still, needs elaborate studies on molecular bioremediation in several metal areas worldwide specifically using fish and parasites.

Procedure for bioaccumulation studies using fish and their parasites

Agricultural waste, industrial waste, and mining waste are the major sources of aquatic metal pollution apart from domestic waste (Brazova et al. 2012). These metals in water bodies will enter into fish and their parasites and affect their growth and development. To do monitoring of metals using fishes and their parasites, the following procedure was followed by several workers (Keke et al. 2020). Initially, the water bodies near various metal-rich areas need to be assessed to know the type of metals and their concentrations through atomic absorption spectrophotometer (AAS) or inductively coupled plasma optical emission spectrometer (ICP-OES). In addition, a detailed ichthyofaunal survey along with their parasites in the metal-rich areas will be carried out (Amarnath Reddy et al. 2013). Fish and their parasite selection for biomonitoring work will be based on the presence and dominance of the species.

It is important to know whether the selected fish species inhabit the water bodies of metal-rich and normal areas. Initially, fish samples and their parasites are to be collected from both metal-rich and normal water

bodies. After the collection of fish samples, parasites will be carefully detached as per the procedure of Vankara and Chikkam (2017). Morphological, physiological, and biochemical studies will be conducted to know their growth and development under metal stress (Chadamala and Vankara 2018). Later, fish muscle and parasite tissue samples will be analyzed to know the metal accumulation capacity and lethal doses. Specifically, fish muscle (150 mg) and parasite tissue (50 mg) samples will be treated with the solution containing a mixture of 2 mL HNO_3 (65%, suprapure) and 2.5 mL H_2O_2 (30%, suprapure) and heated for 90 min at approximately 170°C in a microwave digestion system (Keke et al. 2020). After digestion, the samples will be diluted with deionized water and used for metal analysis using AAS or ICP-OES. Further, estimation of the lethal dose needs to be carried out using more samples. Based on the content, it is easy to identify the fish and their parasites as indicators of metals.

Future prospects

It is a well-known fact that metals are one of the important environmental pollutants which cause potential risk for aquatic fauna including fish and parasites in turn for human beings. We cannot avoid the natural reasons i.e. weathering and volcanic eruption, but anthropogenic reasons are under our control. Domestic waste must be properly collected and disposed in dumping areas only. Dumping waste must be treated properly and possibly may reuse. Proper guidelines must be implemented to control the agricultural runoff into water bodies. Industrial and mining waste must be disposed in particular locations instead of disposing into water bodies. Biomonitoring and accumulation studies using plants, algae, and animals is a general process. To investigate the metal contents, different water bodies near agricultural fields, industrial areas, and mining areas are the best places for accurate assessments. Estimation of the accurate existence of the metals is the key factor for future remediation experiments. Moreover, analysis of metals in water bodies using the appropriate advanced tools is a prerequisite for further works.

The collection of fish and parasite samplings is crucial for monitoring and accumulation experiments. Age, sex, and area of collection of fishes and parasites are the key factors for the accumulation experiments. Observation of morphological features of both fish and parasites under metal stress using advanced microscopic techniques is one of the major concerns. Appropriate histopathological and physiological parameters for both fish and their parasites under metal stress are necessary to check the growth and damage. Advanced biochemical and molecular analyses are essential to know the level of metal accumulation, lethal dose and

also know the biochemical changes due to metal stress. Moreover, the metal treatment also causes genotoxicity in various fish and parasites. Hence, using advanced biotechnological tools such as next generation sequencing (NGS) and genome editing (CRISPR-cas9) will be useful to reduce metal toxicity. Overall, fish yield need to be counted using appropriate controls. More samplings will decide the accurate bioindicators. Statistical analysis is a very important stage apart from the fish sampling collection part. The application of advanced metal management practices will improve the fish yield.

Conclusions

Studies on biomonitoring of metal accumulation provide insights into fish and their parasite's growth and development including fish yield. To date, most of the biomonitoring of metal research deals with plants including algal species and less work has been carried out with fish and their parasites. In this review, we emphasized the necessity of metal monitoring in fish and their parasites and sources of metals. The lethal doses of metals for certain fishes and their parasites were highlighted. In addition, morphological, behavioral, histopathological, and biochemical modifications in fish and their parasites due to metal stress were emphasized. The procedure of monitoring metal using fish and their parasites were explained in detail which in turn will be useful for the identification of bioindicators. Overall, fish yield is one of the most concerns and this information may be helpful to avoid the penalty. This updated knowledge can permit the establishment of an effective and predictable bioremediation process in near future.

Author's contribution

VAP and TCS designed the concept and wrote the manuscript. VAP, KR, DVL, and TCS improved the manuscript.

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Data availability statement

All the data included in this manuscript were available from VAP and TCS on reasonable request.

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